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APPLIED PROBABILITY LABORATORY
(22MDC25)

PROJECT 7

**A MINI PROJECT ON APPLICATION OF
JOINT PROBABILITY IN ANALYSING
WEATHER DATA**

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Task 1

1. Joint Probability Mass Function (Joint PMF)

The joint PMF describes the probability that two discrete random variables X and Y take on specific values at the same time. It gives a complete description of the relationship between the two variables.

$$p(x, y) = P(X = x \text{ and } Y = y)$$

$$p(x, y) \geq 0 \text{ for all values of } x \text{ and } y$$

$$\sum_x \sum_y p(x, y) = 1 \rightarrow (\text{Total probability is } 1)$$

2. Marginal Probability Mass Function (Marginal PMF)

The marginal PMF gives the probability of one variable regardless of the other. It is obtained by summing the joint PMF over the other variable. This helps in understanding the behavior of a single variable when you have a joint distribution.

For variable X :

$$p_X(x) = \sum_y p(x, y)$$

For variable Y :

$$p_Y(y) = \sum_x p(x, y)$$

3. Conditional Expectation

The conditional expectation of X given $Y=y$ is the average value of X , when we know that Y has a specific value y . It is a useful tool in probability and statistics to study how the expected value of one variable depends on another.

$$E[X \mid Y = y] = \sum_x x * P(X = x \mid Y = y)$$

$$\text{where } P(X = x \mid Y = y) = P(X = x \text{ and } Y = y) / P(Y = y)$$

This is the conditional probability of X given $Y=y$, assuming $P(Y=y) > 0$

4. Conditional Variance

The conditional variance of X given $Y=y$ measures the spread or variability of X around its

conditional mean, $E[X \mid Y=y]$. It tells us how much X fluctuates when Y is fixed at a certain value.

$$\begin{aligned} \text{Var}(X \mid Y = y) &= E[(X - E[X \mid Y = y])^2 \mid Y = y] \\ &= \sum_x (x - E[X \mid Y = y])^2 * P(X = x \mid Y = y) \end{aligned}$$

Task 2

Go to Web Site http://www.indiawaterportal.org/met_data/.

Download following data

- 1. Monthly rainfall data of Thanjavur District from 1901 to 2002**
- 2. Monthly average maximum temperature at Thanjavur from 1901 to 2002**
- 3. Monthly average minimum temperature at Thanjavur from 1901 to 2002**

Write a Script file in R to identify the month at which

- 1. rainfall and maximum temperature are correlated to the maximum, that is correlation coefficient is maximum**
- 2. rainfall and minimum temperature are correlated to the maximum, that is correlation coefficient is maximum**
- 3. maximum temperature and minimum temperature are correlated to the maximum, that is correlation coefficient is maximum**

Source Code

```
# Load the datasets using base R
rainfall <- read.csv("C:\\R_Project\\rainfall.csv", header = TRUE)
max_temp <- read.csv("C:\\R_Project\\max_temp.csv", header = TRUE)
min_temp <- read.csv("C:\\R_Project\\min_temp.csv", header = TRUE)

# Remove the first column if it's 'Year' to keep only monthly data
rainfall_data <- rainfall[, -1]
max_temp_data <- max_temp[, -1]
min_temp_data <- min_temp[, -1]

# Get month names (column names)
months <- colnames(rainfall_data)

# Initialize vectors for storing correlation values
cor_rain_max <- numeric(length(months))
cor_rain_min <- numeric(length(months))
cor_max_min <- numeric(length(months))
```

```

# Compute correlations month by month
for (i in seq_along(months)) {
  cor_rain_max[i] <- cor(rainfall_data[[i]], max_temp_data[[i]], use =
"complete.obs")
  cor_rain_min[i] <- cor(rainfall_data[[i]], min_temp_data[[i]], use =
"complete.obs")
  cor_max_min[i] <- cor(max_temp_data[[i]], min_temp_data[[i]], use =
"complete.obs")
}

# Find the month with maximum correlation
max_cor_rain_max <- months[which.max(cor_rain_max)]
max_cor_rain_min <- months[which.max(cor_rain_min)]
max_cor_max_min <- months[which.max(cor_max_min)]

# Print results
cat("Month with highest correlation between Rainfall and Max Temp:",
max_cor_rain_max, "\n")
cat("Month with highest correlation between Rainfall and Min Temp:",
max_cor_rain_min, "\n")
cat("Month with highest correlation between Max Temp and Min Temp:",
max_cor_max_min, "\n")

```

Output

```

> # Load the datasets using base R
> rainfall <- read.csv("C:\\R_Project\\rainfall.csv", header = TRUE)
> max_temp <- read.csv("C:\\R_Project\\max_temp.csv", header = TRUE)
> min_temp <- read.csv("C:\\R_Project\\min_temp.csv", header = TRUE)
>
> # Remove the first column if it's 'Year' to keep only monthly data
> rainfall_data <- rainfall[, -1]
> max_temp_data <- max_temp[, -1]
> min_temp_data <- min_temp[, -1]
>
> # Get month names (column names)
> months <- colnames(rainfall_data)
>
> # Initialize vectors for storing correlation values
> cor_rain_max <- numeric(length(months))
> cor_rain_min <- numeric(length(months))
> cor_max_min <- numeric(length(months))
>
> # Compute correlations month by month
> for (i in seq_along(months)) {
+   cor_rain_max[i] <- cor(rainfall_data[[i]], max_temp_data[[i]], use = "complete.obs")
+   cor_rain_min[i] <- cor(rainfall_data[[i]], min_temp_data[[i]], use = "complete.obs")
+   cor_max_min[i] <- cor(max_temp_data[[i]], min_temp_data[[i]], use = "complete.obs")
+ }
>
> # Find the month with maximum correlation
> max_cor_rain_max <- months[which.max(cor_rain_max)]
> max_cor_rain_min <- months[which.max(cor_rain_min)]
> max_cor_max_min <- months[which.max(cor_max_min)]
>
> # Print results
> cat("Month with highest correlation between Rainfall and Max Temp:", max_cor_rain_max, "\n")
Month with highest correlation between Rainfall and Max Temp: Apr
> cat("Month with highest correlation between Rainfall and Min Temp:", max_cor_rain_min, "\n")
Month with highest correlation between Rainfall and Min Temp: Nov
> cat("Month with highest correlation between Max Temp and Min Temp:", max_cor_max_min, "\n")
Month with highest correlation between Max Temp and Min Temp: Nov
> |

```

Task 3

Write a script file in R to form a discrete probability distribution between :

- 1. rainfall and max temperature at the month where correlation coefficient is maximum**
- 2. rainfall and min temperature at the month where correlation coefficient is maximum**
- 3. max temperature and min temperature at the month where correlation coefficient is maximum.**

[The joint probabilities can be worked out by dividing the data on X and Y into different bins (about 5 to 7) and then finding the frequencies in each cell]

- 4. Compute the marginal p.m.fs, conditional expectations and conditional variances.**

Source Code

```
# Load datasets
rainfall <- read.csv("rainfall_thanjavur_1901_2002.csv")
max_temp <- read.csv("max_temperature_thanjavur_1901_2002.csv")
min_temp <- read.csv("min_temperature_thanjavur_1901_2002.csv")

# Remove Year column
rainfall_data <- rainfall[, -1]
max_temp_data <- max_temp[, -1]
min_temp_data <- min_temp[, -1]

# Function to find month with highest correlation
get_max_corr_month <- function(df1, df2) {
  cors <- sapply(1:12, function(i) cor(df1[[i]], df2[[i]], use =
"complete.obs"))
  return(which.max(cors)) # returns month index (1-12)
}

# Find months with highest correlation
month_rm <- get_max_corr_month(rainfall_data, max_temp_data)
month_rn <- get_max_corr_month(rainfall_data, min_temp_data)
month_mn <- get_max_corr_month(max_temp_data, min_temp_data)
```

Discretize into bins

```
get_joint_distribution <- function(x, y, bins = 5) {  
  x_cut <- cut(x, breaks = bins)  
  y_cut <- cut(y, breaks = bins)  
  joint_table <- table(x_cut, y_cut)  
  joint_prob <- prop.table(joint_table) # joint probability matrix  
  return(joint_prob)  
}
```

Compute marginal pmfs

```
get_marginals <- function(joint_prob) {  
  x_marginal <- rowSums(joint_prob)  
  y_marginal <- colSums(joint_prob)  
  return(list(x = x_marginal, y = y_marginal))  
}
```

Compute conditional expectation and variance of Y given X

```
get_conditional_stats <- function(joint_prob) {  
  x_levels <- rownames(joint_prob)  
  y_levels <- colnames(joint_prob)
```

Midpoints of bins (approximated)

```
x_vals <- sapply(strsplit(x_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))  
y_vals <- sapply(strsplit(y_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))
```

```
cond_exp <- numeric(length(x_vals))
```

```
cond_var <- numeric(length(x_vals))
```

```
for (i in 1:length(x_vals)) {  
  row_probs <- joint_prob[i, ]  
  total_p <- sum(row_probs)  
  if (total_p == 0) {  
    cond_exp[i] <- NA  
    cond_var[i] <- NA  
  }
```

```

else {
  cond_y <- as.numeric(y_vals)
  mean_y <- sum(row_probs * cond_y) / total_p
  var_y <- sum(row_probs * (cond_y - mean_y)^2) / total_p
  cond_exp[i] <- mean_y
  cond_var[i] <- var_y
}
}
return(data.frame(X_bin = x_levels, E_Y_given_X = cond_exp,
Var_Y_given_X = cond_var))
}

```

```

# Analysis for Rainfall vs Max Temp
cat("\n=== Rainfall vs Max Temp (Month:", month_rm, ")\n")
joint_rm <- get_joint_distribution(rainfall_data[[month_rm]],
max_temp_data[[month_rm]])
marginals_rm <- get_marginals(joint_rm)
stats_rm <- get_conditional_stats(joint_rm)
print(stats_rm)

```

```

# Analysis for Rainfall vs Min Temp
cat("\n=== Rainfall vs Min Temp (Month:", month_rn, ")\n")
joint_rn <- get_joint_distribution(rainfall_data[[month_rn]],
min_temp_data[[month_rn]])
marginals_rn <- get_marginals(joint_rn)
stats_rn <- get_conditional_stats(joint_rn)
print(stats_rn)

```

```

# Analysis for Max Temp vs Min Temp
cat("\n=== Max Temp vs Min Temp (Month:", month_mn, ")\n")
joint_mn <- get_joint_distribution(max_temp_data[[month_mn]],
min_temp_data[[month_mn]])
marginals_mn <- get_marginals(joint_mn)
stats_mn <- get_conditional_stats(joint_mn)
print(stats_mn)

```

Output

```
> # Load datasets
> rainfall <- read.csv("C:\\R_Project\\rainfall.csv")
> max_temp <- read.csv("C:\\R_Project\\max_temp.csv")
> min_temp <- read.csv("C:\\R_Project\\min_temp.csv")
>
> # Remove Year column
> rainfall_data <- rainfall[, -1]
> max_temp_data <- max_temp[, -1]
> min_temp_data <- min_temp[, -1]
>
> # Function to find month with highest correlation
> get_max_corr_month <- function(df1, df2) {
+   cors <- sapply(1:12, function(i) cor(df1[[i]], df2[[i]], use = "complete.obs"))
+   return(which.max(cors)) # returns month index (1-12)
+ }
>
> # Find months with highest correlation
> month_rm <- get_max_corr_month(rainfall_data, max_temp_data)
> month_rn <- get_max_corr_month(rainfall_data, min_temp_data)
> month_mn <- get_max_corr_month(max_temp_data, min_temp_data)
>
> # Discretize into bins
> get_joint_distribution <- function(x, y, bins = 5) {
+   x_cut <- cut(x, breaks = bins)
+   y_cut <- cut(y, breaks = bins)
+   joint_table <- table(x_cut, y_cut)
+   joint_prob <- prop.table(joint_table) # joint probability matrix
+   return(joint_prob)
+ }
>
> # Compute marginal pmfs
> get_marginals <- function(joint_prob) {
+   x_marginal <- rowSums(joint_prob)
+   y_marginal <- colSums(joint_prob)
+   return(list(x = x_marginal, y = y_marginal))
+ }
>
> # Compute conditional expectation and variance of Y given X
> get_conditional_stats <- function(joint_prob) {
+   x_levels <- rownames(joint_prob)
+   y_levels <- colnames(joint_prob)
+
+   # Midpoints of bins (approximated)
+   x_vals <- sapply(strsplit(x_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))
+   y_vals <- sapply(strsplit(y_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))
+
+   cond_exp <- numeric(length(x_vals))
+   cond_var <- numeric(length(x_vals))
+
+   for (i in 1:length(x_vals)) {
+     row_probs <- joint_prob[i, ]
+     total_p <- sum(row_probs)
+     if (total_p == 0) {
+       cond_exp[i] <- NA
+       cond_var[i] <- NA
+     } else {
+       cond_y <- as.numeric(y_vals)
+       mean_y <- sum(row_probs * cond_y) / total_p
+       var_y <- sum(row_probs * (cond_y - mean_y)^2) / total_p
+       cond_exp[i] <- mean_y
+       cond_var[i] <- var_y
+     }
+   }
+   return(data.frame(X_bin = x_levels, E_Y_given_X = cond_exp, Var_Y_given_X = cond_var))
+ }
>
> # Analysis for Rainfall vs Max Temp
> cat("\n=== Rainfall vs Max Temp (Month:", month_rm, ")\n")

=== Rainfall vs Max Temp (Month: 4 )
> joint_rm <- get_joint_distribution(rainfall_data[[month_rm]], max_temp_data[[month_rm]])
> marginals_rm <- get_marginals(joint_rm)
> stats_rm <- get_conditional_stats(joint_rm)
> print(stats_rm)
      X_bin E_Y_given_X Var_Y_given_X
1 (3.91,82.7]    34.79800    10.469496
2 (82.7,161]    37.33235     9.036453
3 (161,239]    36.23056    11.740594
4 (239,318]    35.65714     9.092925
5 (318,397]    35.76905     8.435113
>
```

```

> # Analysis for Rainfall vs Min Temp
> cat("\n=== Rainfall vs Min Temp (Month:", month_rn, ")\n")

=== Rainfall vs Min Temp (Month: 11 )
> joint_rn <- get_joint_distribution(rainfall_data[[month_rn]], min_temp_data[[month_rn]])
> marginals_rn <- get_marginals(joint_rn)
> stats_rn <- get_conditional_stats(joint_rn)
> print(stats_rn)
      X_bin E_Y_given_X Var_Y_given_X
1 (6.71,85.5]    22.45000    5.070000
2 (85.5,164]    21.88125    8.813086
3 (164,242]    22.96071   10.030064
4 (242,321]    23.51053    9.037521
5 (321,400]    23.60556    7.771914
>
> # Analysis for Max Temp vs Min Temp
> cat("\n=== Max Temp vs Min Temp (Month:", month_mn, ")\n")

=== Max Temp vs Min Temp (Month: 11 )
> joint_mn <- get_joint_distribution(max_temp_data[[month_mn]], min_temp_data[[month_mn]])
> marginals_mn <- get_marginals(joint_mn)
> stats_mn <- get_conditional_stats(joint_mn)
> print(stats_mn)
      X_bin E_Y_given_X Var_Y_given_X
1 (30,32.4]    22.07857    7.467041
2 (32.4,34.8]    22.91429   10.467653
3 (34.8,37.1]    22.65682    9.309840
4 (37.1,39.5]    23.56429    6.846224
5 (39.5,41.9]    23.32941    7.105017
> |

```